

Calculus & Linear Algebra for Computer Scientists

Author: Thomas H. Cormen | Category: Mathematics

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Unit 1: Vectors & Matrix Operations

Vector spaces, dot products, cross products, and geometric projections.

Matrix multiplication, determinants, and Gaussian elimination methods.

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Unit 2: Eigenvalues and Eigenvectors

Characteristic polynomials and diagonalizing square matrices.

Principal Component Analysis (PCA) and dimensionality reduction.

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Unit 3: Differential Calculus in Multiple Dimensions

Partial derivatives, directional derivatives, and gradient vectors.

Jacobian and Hessian matrices for multivariable functions.

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Unit 4: Optimization & Gradient Descent

Convex functions, saddle points, and local vs global extrema.

Stochastic Gradient Descent (SGD) with momentum in neural network training.

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Unit 5: Integrals & Probability Densities

Double and triple integrals for area and volume calculations.

Joint probability distributions, expectation, and variance formulations.